

Does the Value of Residual Connections Depend on Depth?

A Parameter-Matched, Multi-Seed Ablation at 8 and 20 Layers on CIFAR-10

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Code and raw results: <https://github.com/AlanPai777/shallow-resnet>

Abstract

A residual (skip) connection lets a layer's input bypass the layer and be added back to its output. It is usually credited with making very deep networks trainable, so I wanted to know whether it still helps in a shallow one. I compared networks with and without skip connections at 8 and 20 layers on CIFAR-10. The two versions have exactly the same number of parameters, because the shortcuts contain no weights, and batch normalization is on in both. Each configuration was trained with three random seeds for 10 epochs.

With an annealed learning rate and no checkpoint selection, removing the skip connections made no detectable difference at 8 layers (-0.42 points, 95% CI $[-3.22, +2.38]$) and cost 2.48 points at 20 layers (95% CI $[+1.73, +3.24]$). The difference between the two depths is statistically significant ($p = 0.005$, or 0.036 under a more conservative test). At 20 layers the gap is already present in training loss, so it comes from easier optimization rather than better generalization. It did not shrink over the last epochs. The result covers two depths and a short training budget, so it does not locate a threshold depth.

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1 Introduction

1.1 Where the question came from

In a course assignment on CIFAR-10 I trained two CNNs. One had six convolution layers with batch normalization (BN) and no skip connections (1.66M parameters). The other had three residual blocks, also with BN (about 2.86M parameters). They reached almost the same test accuracy: 85.08% and 84.88%, from one run each.

That comparison says very little about skip connections. The two networks differ in width, in block structure, and in parameter count (the residual one also uses 1×1 projection shortcuts, which add parameters). A 0.2-point gap between single runs is also within what random initialization alone can produce. But it left me with a question I wanted to answer properly: if everything else is held fixed, does a skip connection help, and does the answer depend on how deep the network is?

1.2 Why the question is not settled by the original ResNet paper

Skip connections were introduced to fix the degradation problem: past a certain depth, plain networks get higher training error than shallower ones (He et al., 2016a). In that paper, as in most work that followed, the plain baseline already contains BN (Ioffe & Szegedy, 2015). So what gets measured is the value of a skip connection given BN, and the two techniques are rarely separated.

He et al. compare plain and residual networks on CIFAR-10 from 20 layers upward, with matched parameter counts. I did not find a replicated comparison below 20 layers. There is also a reason to expect a small effect in shallow networks: Veit et al. (2016) show that a ResNet behaves like a collection of paths, most of them much shorter than the nominal depth. If the network is already short, the skip connection has little to shorten.

1.3 What I did

I built a pair of networks that differ only in whether the shortcut is present, using the CIFAR-10 ResNet layout of He et al. (2016a), and trained both at 8 and 20 layers with three seeds each. I ran the experiment in two stages, because the first turned out to have a flaw:

1. **Stage 1** used the recipe from my course code: constant learning rate, and the reported accuracy is that of the epoch with the lowest loss on the test set. This makes the reported number depend on the test set. Once I looked at the results I also found that, with a constant learning rate, the last-epoch accuracy was too noisy to compare.
2. **Stage 2** repeated all twelve runs with a cosine learning-rate schedule and reported the accuracy after the last epoch, with no selection. This is the primary result.

Both stages are reported, because the difference between them shows how sensitive this kind of comparison is to the evaluation protocol.

1.4 Findings in brief

- **Depth 8:** no detectable effect of the skip connection (-0.42 points, $p = 0.61$). The interval is wide, so effects of two to three points in either direction are not excluded.
- **Depth 20:** the skip connection helps by 2.48 points, with all three seeds with the skip connection above all three without it.
- **Depth-dependence:** the interaction between depth and skip connection is significant.
- **Protocol:** the depth-20 estimate varies from $+1.90$ to $+3.26$ points depending on how accuracy is reported, and only the annealed, no-selection protocol gives an interval narrow enough to interpret.
- **Mechanism:** the depth-20 gap persists to the last epoch and shows up in training loss. Neither network shows vanishing gradients in early layers.

2 Background

Residual networks and BN. A residual block computes $y = F(x) + x$, where F is a few convolution layers. He et al. (2016a) argued that deep plain networks are hard to optimize, not that they overfit, and showed that their plain networks (with BN) keep healthy gradient norms. They suggested the cause is slow convergence. BN itself helps mainly by making the loss surface smoother rather than by reducing "internal covariate shift" (Santurkar et al., 2018).

Why a shallow network might not need skips. Veit et al. (2016) show that a ResNet acts like an ensemble of paths of many different lengths, and that in a 110-layer network the gradient during training comes almost entirely from short paths. For a network that is short to begin with, this suggests the skip connection changes little.

Skip-connection ablations at other depths. He et al. (2016a) compared plain and residual networks on CIFAR-10 starting at 20 layers, shown as training and test curves. A short arXiv report (Liu & Goh, 2025) removes the skips from an 18-layer ResNet on CIFAR-10 and reports 89.9% versus 86.0% test accuracy, apparently from one run per model. My design follows the parameter-free shortcut of He et al. and adds a shallower depth and repeated seeds.

3 Method

The only intended difference between the two networks in each pair is the shortcut.

3.1 Terminology

Term	Meaning in this report
Skip connection / shortcut	The path that adds a block's input to its output.
Residual ON / OFF	The network with / without the shortcuts. "OFF" is a plain network; BN stays on in both.
Depth	$6n + 2$ layers: one stem convolution, three stages of n blocks with two convolutions each, one fully connected layer.
Cell	One combination of depth and residual ON/OFF (four cells in total).
Seed	The random seed fixing initialization and data order. Each cell has three seeds: 2026, 2027, 2028.
Effect of the skip connection	Mean accuracy with skips minus mean accuracy without, in percentage points.
Interaction	Whether that effect differs between depth 8 and depth 20 (a difference of two differences).
95% CI	95% confidence interval for a difference of means.

3.2 Networks

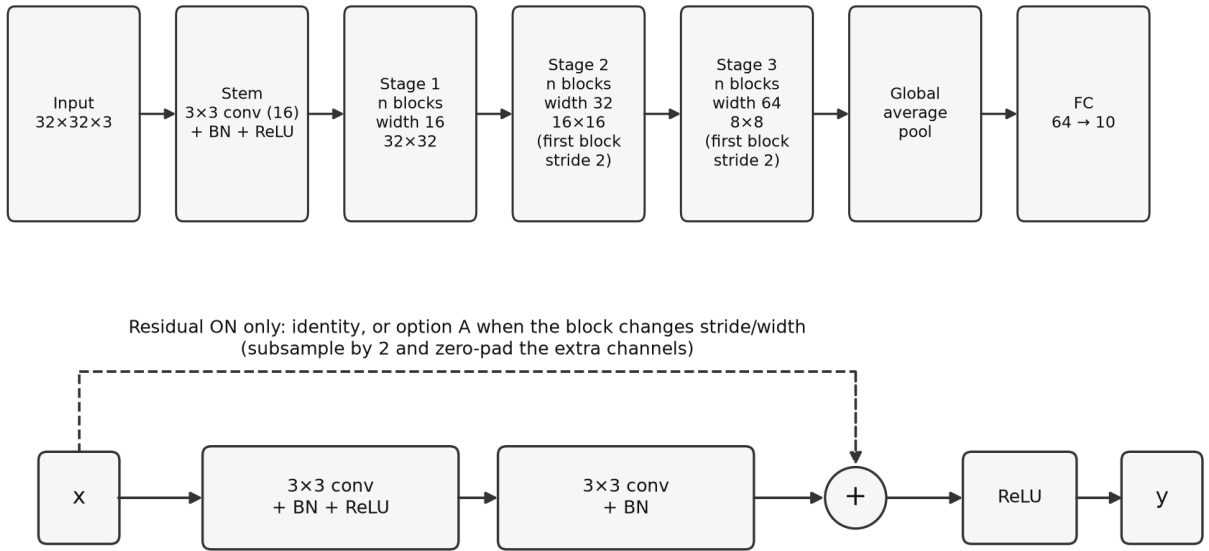


Figure 1. The network (top) and one block (bottom). In the residual-OFF network the dashed path and the addition are removed and nothing else changes.

- **Shortcuts have no parameters.** When a block keeps its width and stride, the shortcut is the identity. When a block doubles the width and halves the resolution, the shortcut subsamples the input with stride 2 and pads the extra channels with zeros (called "option A" by He et al., 2016a). So the ON and OFF networks have exactly the same parameters: 75,290 at depth 8 ($n = 1$) and 269,722 at depth 20 ($n = 3$). The second matches the 0.27M reported for ResNet-20 by He et al. I check the equality in code.
- **BN is on in every run**, so the comparison is "skip connection given BN". This report says nothing about networks without BN.
- **Initialization** is He normal in fan-out mode (He et al., 2015); BN scale and shift start at one and zero; convolutions have no bias.

3.3 Training setup

CIFAR-10 (Krizhevsky, 2009) with the standard 50,000/10,000 train/test split. Per-channel mean and standard-deviation normalization; for training, random 32×32 crops from images padded by 4 pixels and random horizontal flips. Adam (Kingma & Ba, 2015) with initial learning rate 0.001, weight decay $1e-4$ applied as L2 regularization, batch size 64, 10 epochs, cross-entropy loss, no dropout. Runs were done in PyTorch on a single Google Colab GPU. Each run fixes the Python, NumPy and PyTorch seeds and seeds the data loader, so the seed determines both initialization and the order of examples. cuDNN is set to deterministic mode.

The full design is 24 runs: 2 depths $\times$ 2 (skip ON/OFF) $\times$ 3 seeds $\times$ 2 learning-rate recipes. I originally planned a larger study (BN on/off, five depths, several batch sizes, five or more seeds, roughly a hundred runs), and this is the part that fit the compute available. It leaves out the BN-off arm, intermediate depths, and larger seed counts. Section 5 discusses what that costs.

3.4 Two learning-rate recipes and how accuracy is reported

	Learning rate	Reported accuracy
Constant (stage 1)	$1e-3$ throughout	Either the epoch with the lowest test loss (inherited from my course code; called the legacy metric below), or the final epoch
Cosine (stage 2, primary)	Cosine annealing without restarts (Loshchilov & Hutter, 2017), period 10 epochs, stepped once per epoch: $1.0e-3$, $9.8e-4$, $9.0e-4$, $7.9e-4$, $6.5e-4$, $5.0e-4$, $3.5e-4$, $2.1e-4$, $9.5e-5$, $2.4e-5$	Final epoch

Three things about the evaluation matter for reading the numbers:

- **There is no separate validation set.** Every epoch, the model is evaluated on the 10,000 test images. In the legacy metric that evaluation also chooses the epoch, so the reported accuracy is optimistic. In this report the per-epoch "validation" loss and accuracy are computed on the test set, and I call them "test" from here on except in figure captions. Under the primary protocol no decision depends on the test set (the per-epoch values are only logged for the curves).
- **The hyperparameters were inherited.** Learning rate, weight decay and epoch count come from the course assignment and were not tuned on held-out data, nor separately per arm. Choosing to try a cosine schedule came after I had seen the test curves of stage 1. Only these two recipes were run.
- **Accuracy and loss are averages of per-batch values** over the 157 evaluation batches (the last has 16 images). This weights the last batch slightly more than a count over all images would, by at most a few tenths of a point, and the same way for every run.

3.5 Statistics

I report means and sample standard deviations (denominator $n - 1$). Simple effects use Welch's t-test and an exact permutation test; with three versus three observations the smallest possible two-sided permutation p is 0.10, so that test cannot reach conventional significance and I use it only as a consistency check. The interaction is tested with a two-way ANOVA on the balanced 2×2 design (8 residual degrees of freedom). Because ANOVA

assumes equal variances and these cells clearly differ (variances from 0.06 to 1.45), I also report an unpooled (Welch-Satterthwaite) test of the difference-in-differences.

3.6 Gradient diagnostic

For seed 2026 only, at each depth and for both networks, I recorded the L2 norm of the gradient of every convolution weight tensor, averaged over the first 50 optimizer steps (Adam, learning rate 1e-3, no weight decay in this diagnostic). It is a lightweight check on whether early layers are starved of gradient, not a full analysis of optimization.

4 Results

4.1 Main result

Table 1. Final-epoch test accuracy (%), cosine schedule (primary protocol).

Depth	Skip connection	Seed 2026	Seed 2027	Seed 2028	Mean	SD (n – 1)
8	ON	75.26	74.87	73.01	74.38	1.20
8	OFF	75.12	74.63	74.64	74.80	0.28
20	ON	82.36	82.71	82.81	82.63	0.24
20	OFF	79.72	80.33	80.38	80.15	0.37

Depth 8. With skips is 0.42 points below without, 95% CI [–3.22, +2.38] (Welch $t = -0.59$, $p = 0.61$; permutation $p = 0.90$). There is no detectable effect. The interval is wide, mostly because one ON run (seed 2028, 73.01%) is lower than the rest, and it does not rule out effects of two to three points either way.

Depth 20. With skips is 2.48 points above, 95% CI [+1.73, +3.24] (Welch $t = 9.79$, $p = 0.0013$; permutation $p = 0.10$, the smallest value possible). All three ON seeds (82.36 to 82.81) beat all three OFF seeds (79.72 to 80.38).

Interaction. The question is whether the effect depends on depth, so I test the interaction directly instead of comparing two separately tested effects. The difference-in-differences is +2.90 points (pooled 95% CI [+1.16, +4.65]); $F(1, 8) = 14.73$, $p = 0.005$. The pooled test is dominated by the noisy depth-8 ON cell, so I also ran the unpooled test: $t = 3.84$, $p = 0.036$, 95% CI [+0.39, +5.42]. The conclusion that the effect differs between the two depths holds under both, though the interval under the conservative test is wide. (The main effect of skips averaged over depths, $F = 7.47$, $p = 0.026$, comes entirely from depth 20 and should not be read as a general effect.)

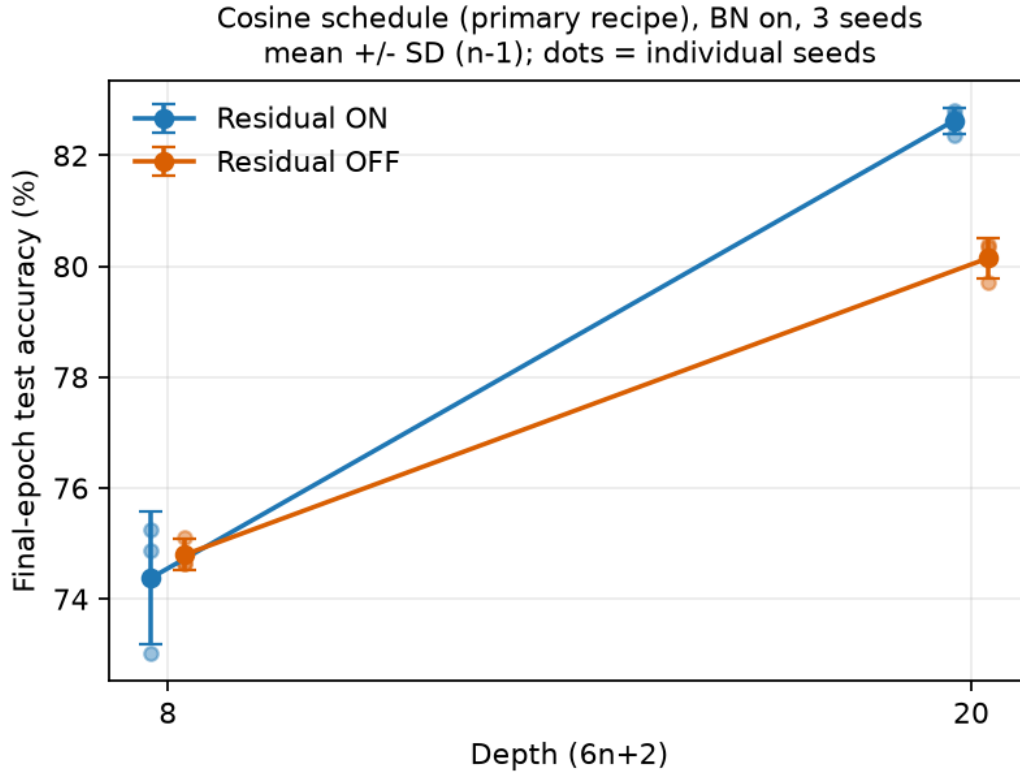


Figure 2. Final-epoch test accuracy by depth and skip connection, cosine schedule, BN on. Error bars: sample SD over three seeds; dots: individual seeds. The lines join two points and say nothing about the shape between them.

4.2 How much the result depends on the evaluation protocol

Table 2. Effect of the skip connection (ON – OFF, points, 95% CI) under three ways of reporting accuracy on the same design.

Protocol	Depth 8	Depth 20	Interaction (diff.-in-diff.; pooled ANOVA p)
Constant LR, epoch with lowest test loss (legacy; uses the test set to choose)	–1.38 [–3.81, +1.04]	+3.26 [+0.81, +5.70]	+4.64; p = 0.002
Constant LR, final epoch (no selection)	–1.03 [–3.23, +1.18]	+1.90 [–2.72, +6.52]	+2.92; p = 0.10
Cosine LR, final epoch (no selection; primary)	–0.42 [–3.22, +2.38]	+2.48 [+1.73, +3.24]	+2.90; p = 0.005

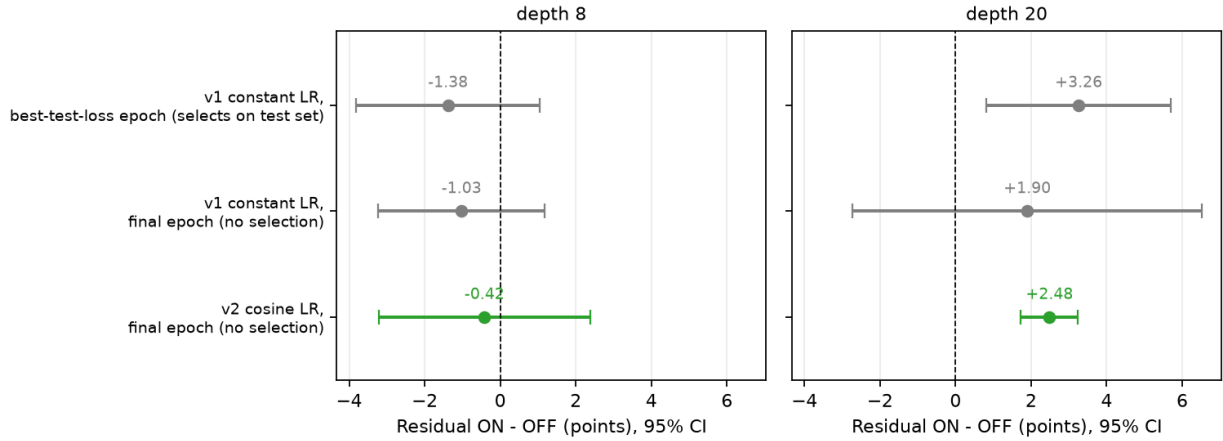


Figure 3. Estimated effect of the skip connection and its 95% CI under the three protocols of Table 2, at depth 8 (left) and depth 20 (right).

- **The legacy number looks best for the skip connection.** It gives +3.26 at depth 20 against +1.90 for the same runs without selection. Nearly all of that 1.4-point difference comes from one run: the depth-20 ON run with seed 2028 has legacy accuracy 81.32% but 77.39% after its final epoch. The other five depth-20 runs differ between the two metrics by at most 0.2 points. With these data I cannot separate the effect of selecting on the test set from the effect of one unlucky last epoch. What I can say is that the legacy number rests on a choice made with the test set, that ON curves at depth 20 are the noisier ones (mean epoch-to-epoch change in test accuracy 4.0 versus 3.4 points), and that noisier curves benefit more from picking their best epoch. The legacy figure should not be quoted alone.
- **Without selection, a constant learning rate is too noisy.** The depth-20 ON cell has a standard deviation of 2.19 points because of that one run, and the interval for the effect runs from -2.72 to $+6.52$.
- **Annealing fixes the noise.** The standard deviations of the depth-20 cells drop from 2.19 and 0.99 to 0.24 and 0.37 points. Under annealing eleven of the twelve runs reach their lowest test loss at the last epoch, so legacy and final-epoch accuracy coincide and the selection problem disappears.

Table 3. Final-epoch accuracy (%), mean $\pm$ SD over three seeds, by learning-rate schedule.

Depth	Skip connection	Constant LR	Cosine LR	Change
8	ON	73.86 $\pm$ 1.04	74.38 $\pm$ 1.20	+0.52
8	OFF	74.88 $\pm$ 0.86	74.80 $\pm$ 0.28	-0.08
20	ON	79.92 $\pm$ 2.19	82.63 $\pm$ 0.24	+2.71
20	OFF	78.02 $\pm$ 0.99	80.15 $\pm$ 0.37	+2.12

Annealing helps the 20-layer networks by about 2 to 2.7 points and the 8-layer networks by almost nothing, which is consistent with the shallow networks being close to converged already at 10 epochs.

4.3 Where the depth-20 advantage comes from

Solid = cosine (mean; band = min-max over 3 seeds); dashed = constant-LR mean

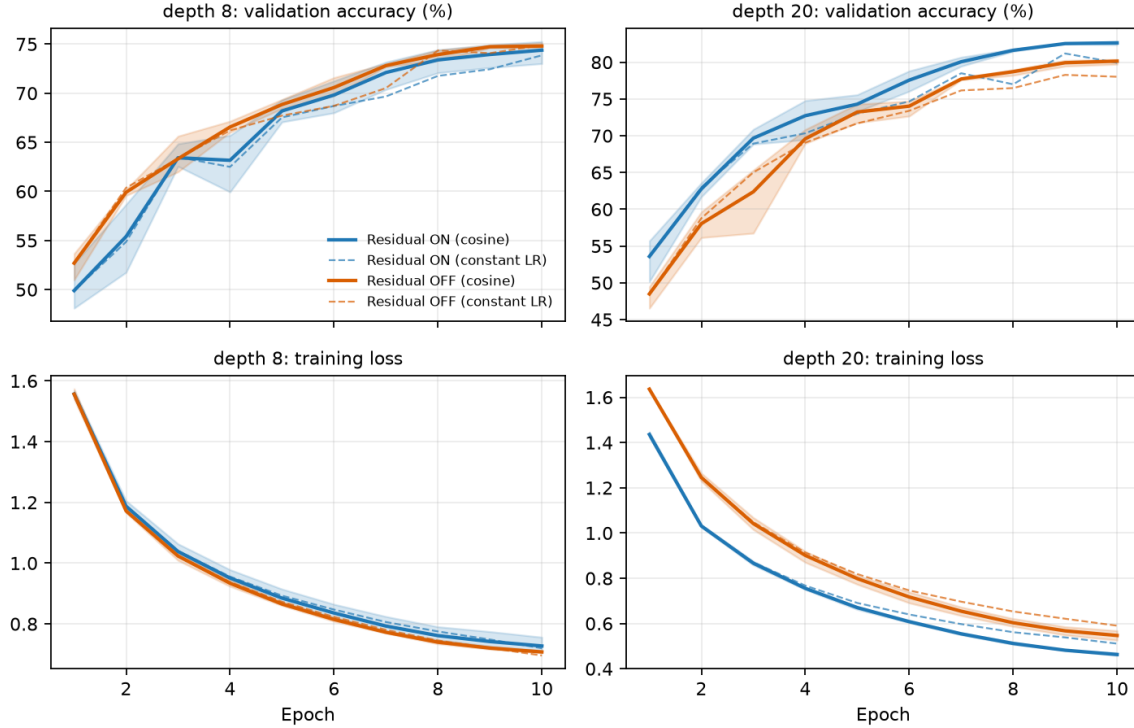


Figure 4. Mean test accuracy (top) and training loss (bottom) per epoch, at depth 8 (left) and depth 20 (right). Solid: cosine runs, bands show min-max over three seeds. Dashed: means under the constant learning rate.

With a constant learning rate the depth-20 gap was large after the first epoch (+5.1 points) and smaller later (0.5 to 2.9 points over the last three epochs). That suggested the skip connection mainly speeds up early training and that the gap might vanish with more training. Under annealing that did not happen. The per-epoch gap in mean test accuracy at depth 20 is +5.1, +4.7, +7.3, +3.1, +1.1, +3.5, +2.3, +2.9, +2.6 and +2.5 points: largest early, then steady at 2.3 to 2.9 points over the last four epochs instead of closing.

The gap also appears in the losses at the end of training. At depth 20, final training loss is 0.463 with skips against 0.547 without (difference -0.084 , 95% CI $[-0.131, -0.037]$, $p = 0.014$), and final test loss is 0.507 against 0.574 (difference -0.068 , 95% CI $[-0.096, -0.039]$, $p = 0.009$). At depth 8 the differences are small and indistinguishable from zero (training loss +0.020, $p = 0.31$; test loss +0.024, $p = 0.33$).

The skip connection does not shrink the gap between training and test accuracy. At the last epoch that gap is 1.4 points with skips and 1.0 without at depth 20, and 0.2 versus 0.5 at depth 8. (Training accuracy is the running average over the epoch, with augmentation and BN in training mode, so it is only comparable between the two networks.) The advantage is therefore about how well the network fits the training data, not about generalizing better.

An unexplained observation at depth 8. The plain network leads by several points in the early epochs under both recipes (-2.8 , -4.5 and -3.4 points in epochs 1, 2 and 4 under cosine), and the gap closes to -0.4 by the last epoch. I did not investigate why.

4.4 Gradient norms across layers

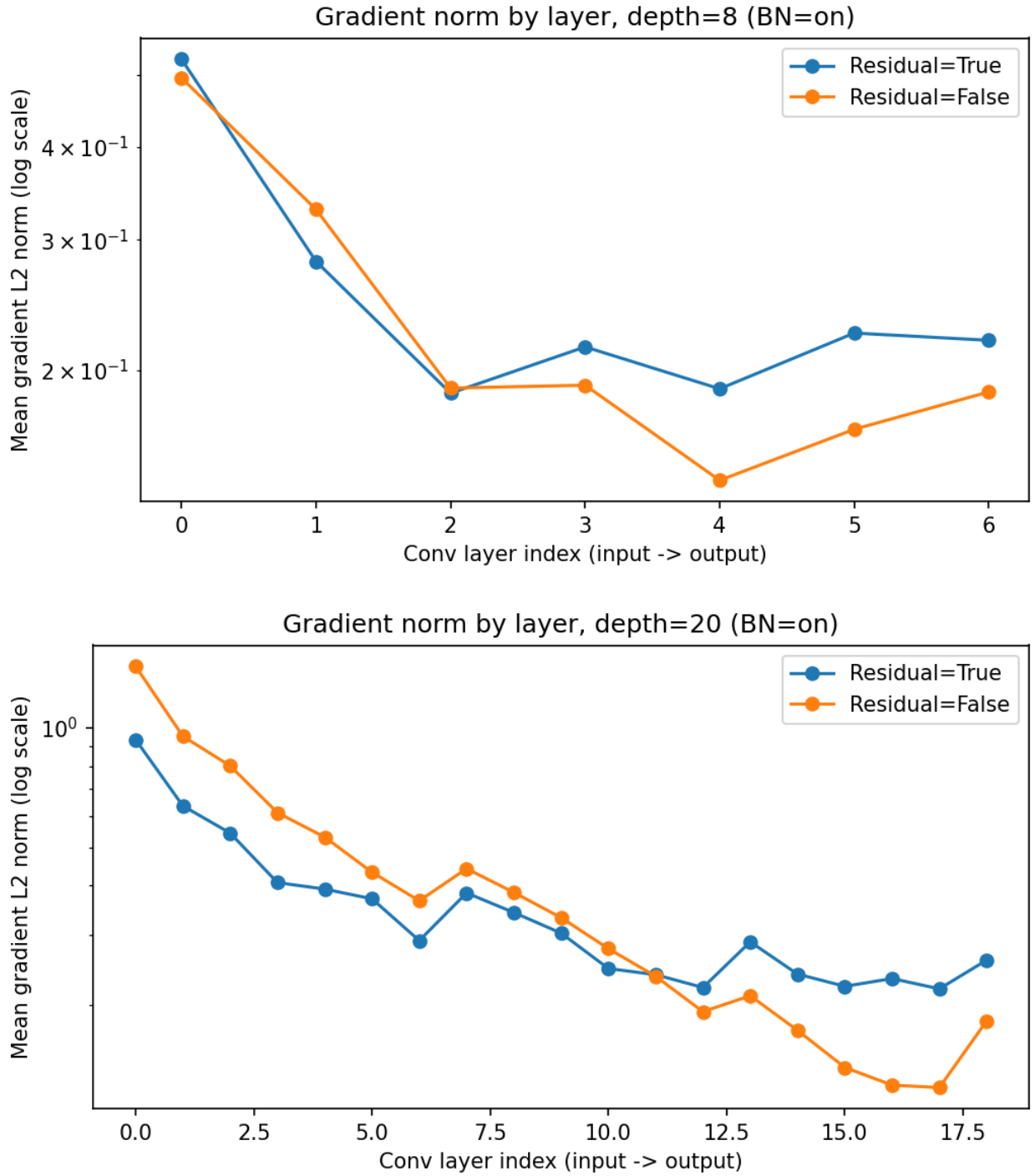


Figure 5. Mean L2 norm of the gradient of each convolution weight tensor (log scale), input on the left, output on the right, for depth 8 (7 convolutions) and depth 20 (19 convolutions). Seed 2026 only, averaged over the first 50 optimizer steps at learning rate 1e-3.

In both networks at both depths the gradient norm is largest near the input and smaller toward the output. Neither network shows early-layer vanishing gradients, so this measurement does not support a vanishing-gradient explanation for the depth-20 gap. The network with skips has a flatter profile. Reading the figures, the ratio between the largest and smallest per-layer norm is roughly 3 (ON) against 4 (OFF) at depth 8, and roughly 4

against 10 at depth 20. These ratios were read off the plots, because I did not save the raw per-layer values.

This is weak evidence. It uses one seed, it averages over the first 50 steps instead of measuring at initialization, it covers only convolution weights, and with BN the norm of a weight's gradient depends inversely on the weight's norm, so raw norms are a rough proxy for how well signal propagates. What it does support is that skip connections make gradient scale more uniform across layers, a little more at depth 20 than at depth 8.

4.5 Sanity checks

- **Parameter counts** are identical between ON and OFF at both depths (75,290 and 269,722).
- **Epoch 1 is identical across the two recipes.** The learning rate is $1e-3$ in the first epoch of both, and the seed fixes initialization and data order, so epoch-1 loss and accuracy must match. Across all twelve matched runs the maximum difference is exactly zero, even though the cosine runs were executed later. This confirms that the two stages differ only from the second epoch onward.
- **Comparison with He et al.** My depth-20 network reaches 82.6% (cosine) against their 91.25% for ResNet-20. That gap is expected: they trained for 64k iterations at batch size 128 (about 180 epochs over a 45k-image training split) with SGD, momentum and step decay, and I trained for 10 epochs with Adam. I treat this only as a plausibility check that the implementation is sound, not a reproduction.

5 Discussion and limitations

What the results support. Under this setup the skip connection helps at 20 layers and makes no detectable difference at 8, and the difference between the two depths is statistically supported. The 20-layer advantage is an optimization advantage: it is visible in training loss and persists through the end of an annealed schedule. It is in the same direction as the 18-layer result of Liu & Goh (2025), 89.9% versus 86.0%, although their network has about forty times more parameters and a different recipe. At 8 layers, the absence of an effect is what the "short effective paths" view of Veit et al. (2016) would lead one to expect, but this is a consistency observation, not a test of that account. He et al. (2016a) report that on ImageNet at 18 layers plain and residual networks end up at similar accuracy; on CIFAR-10 at 20 layers and within my budget they do not, though the datasets, depths and budgets differ.

Limitations.

- **Two depths only.** The data support "the effect differs between 8 and 20 layers", not a claim about where it appears or how it grows. The line in Figure 2 joins two points. Adding depths (with the same replication) is the most direct extension.
- **BN was never switched off.** The design says nothing about how much of the effect depends on BN, or whether BN alone is enough at shallow depth, which was part of my original motivation. Batch size is a related untested axis: BN degrades at small batches.
- **Limited statistical power.** Three seeds per cell give an interval of about ± 2.8 points at depth 8. The smallest attainable permutation p-value is 0.10, the depth-20 t-test rests on about three degrees of freedom, and the interaction p-value ranges from 0.005

to 0.036 depending on whether variances are pooled. Five or more seeds per cell, under both recipes, would tighten this.

- **No held-out validation set.** The primary protocol chooses no epoch, but the hyperparameters were inherited and not tuned on validation data, and the decision to try cosine annealing came after I had seen the stage-1 test curves. It was applied to both arms, and no other recipe was tried. A validation split with the recipe fixed in advance would remove these concerns. The legacy metric should not be quoted alone.
- **Short training.** Ten epochs is about one-eighteenth of He et al.'s schedule, and my depth-20 accuracy is 8.6 points below theirs, so none of the networks is converged. That the gap persists under annealing rules out only the simplest form of "the skip connection only speeds things up early". It does not show that a plain network would never catch up under 30 to 100 epochs or more.
- **A single dataset and a narrow architecture family.** The blocks are post-activation (ReLU after the addition), with identity or option-A shortcuts, widths of 16–64, Adam with coupled L2 decay. In post-activation blocks the identity path passes through a ReLU after each addition, so it is not a clean identity (He et al., 2016b); I did not test whether this matters. Results may not carry over to pre-activation blocks, projection shortcuts, wider networks, SGD, or other datasets such as CIFAR-100.
- **The gradient diagnostic is thin.** One seed, first 50 steps only, convolution weights only, and layer-wise ratios read from the figures. Several seeds, measurements at initialization, and the smoothness measures of Santurkar et al. (2018) would give more.

6 Conclusion

With shortcuts that add no parameters, BN on, and three seeds per configuration, removing skip connections has no detectable effect on CIFAR-10 accuracy at 8 layers (−0.42 points, 95% CI [−3.22, +2.38]) and costs about 2.5 points at 20 layers (95% CI [+1.73, +3.24]), after 10 epochs with an annealed learning rate and no checkpoint selection. The difference between the depths is significant ($p = 0.005$ pooled, 0.036 unpooled). The 20-layer gap appears in training loss and does not close by the end of the schedule. Gradient norms show a flatter profile with skips but no early-layer vanishing in either network.

The same design gave a depth-20 estimate anywhere from +1.90 to +3.26 points depending on how accuracy was reported, so how a comparison like this is evaluated matters as much as the model change being tested. The result concerns two depths and one short training budget. It does not locate a threshold depth and says nothing about networks without BN.

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